STERLING TRUST BANK | BANK LOAN PORTFOLIO & CREDIT RISK ANALYSIS: KEY FINDINGS AND RECOMMENDATIONS

KEY FINDINGS

1. Does credit score affect loan repayment?

Yes, people with credit scores above 700 have a repayment rate of 89%, compared to people with credit scores below 500, who have a repayment rate of 65%, and those with credit scores between 501 and 600, who have a repayment rate of 68%.

2. Does a high debt-to-income ratio increase repayment risk?

Yes, people with a low debt-to-income ratio have about a 91% repayment rate, with 4,894 out of 5,403 loans paid back. In comparison, people with a DTI of 40% and above have a non-repayment rate of 43%, with 404 out of 711 loans paid back.

4. Which employment categories have the highest repayment rate?

Retired customers have the highest repayment rate at 99%, with 1,170 out of 1,176 loans paid back. They are followed by self-employed and employed customers, both with repayment rates of 89%. Employed customers paid back 11,537 out of 13,007 loans, while self-employed customers paid back 2,589 out of 2,923 loans.

5. Does income level appear to affect repayment?

There is not much difference in repayment rates across income levels. All income groups, from those earning below 30k to those earning above 100k, have repayment rates of 79% and above.

6. Which loan purposes have the highest non-repayment rate?

Education and medical loans have the highest non-repayment rates, both at approximately 22%. Education loans had 1,303 paid-back loans out of 1,675 total loans, while medical loans had 936 paid-back loans out of 1,196 total loans.

7. Are larger loans associated with greater repayment risk?

No, not necessarily. Loans above 30k have a repayment rate of 81%, with 753 loans paid back and 180 unpaid out of 933 total loans. People who borrowed below 5k had 2,223 loans paid back and 543 unpaid out of 2,766 total loans, giving a repayment rate of 80%. The difference is small, so it's more accurate to say larger loans do not appear to have a substantially higher repayment risk based on these two groups alone.

8. Does interest rate appear to affect repayment?

Yes, interest rates appear to affect repayment. Loans with interest rates below 5% have a 91% repayment rate. Out of 23 loans, 21 were paid back and only 2 were unpaid.

People with interest rates between 5% and 10% have a repayment rate of 87%, with 2,810 loans paid back out of 3,238 total loans.

However, loans with interest rates of 20% and above have a repayment rate of 67%, the lowest among the interest rate groups. Only 15 loans were granted in this group, with 10 paid back and 5 unpaid.

9. Which credit grades perform best and worst?

Credit Grade A, which includes grades A1–A5, performed best, with a repayment rate of 95%.

Credit Grade F, which includes grades F1–F5, performed worst, with a repayment rate of 68%. Out of 1,551 loans, 1,048 were paid back and 503 were unpaid.

10. Which customer segment represents the highest credit risk?

Based on the table, unemployed borrowers appear to represent a high-risk customer segment. Several combinations of credit score and DTI bands show a 100% non-repayment rate, with zero loans paid back.

For example:

- Credit score 501–600, unemployed, DTI above 10%–20%: 111 loans, 100% non-repayment.
- Credit score 601–655, unemployed, DTI above 10%–20%: 182 loans, 100% non-repayment.
- Credit score 601–655, unemployed, DTI above 20%–30%: 124 loans, 100% non-repayment.

Conclusion: Unemployed borrowers, particularly those in the credit score and DTI groups listed above, represent the highest observed credit risk in this table.

Recommendations for Sterling Trust Bank

1. Improve Loan Approval Decisions

The bank should carefully assess unemployed applicants before approving loans. For example, unemployed borrowers aged 21 had only 5 out of 41 loans paid back, giving a repayment rate of 12%. Similar low repayment rates were observed among unemployed borrowers aged 23 and 24. The bank should review their repayment capacity before approving loans.

2. Strengthen Credit Assessment

The bank should consider credit scores, debt-to-income (DTI) ratios, employment status, and delinquency history when assessing applicants. Customers with credit scores above 700 had an 89% repayment rate, while those with scores below 500 had a 65% repayment rate. Customers with DTI ratios of 40% and above also showed higher repayment risk.

3. Improve Risk Monitoring

The bank should closely monitor high-risk customer groups, especially unemployed borrowers and customers with high DTI ratios. It should identify early signs of repayment difficulties and contact customers who miss payments to reduce further losses.

4. Use Risk-Based Interest Rates

The bank should consider borrowers' credit scores, DTI ratios, and repayment capacity when setting interest rates. Loans with interest rates of 20% and above had a repayment rate of 67%, although this group contained only 15 loans. The bank should investigate whether higher rates are associated with repayment difficulties before making pricing decisions.

5. Manage High-Risk Customer Segments

The bank should pay closer attention to unemployed borrowers and student borrowers, particularly younger applicants. For example, unemployed borrowers aged 22 had a repayment rate of 30%, while students aged 22 had a repayment rate of 35%. In comparison, self-employed borrowers aged 24 had a repayment rate of 93%, and employed borrowers of the same age had a repayment rate of 87%. These differences suggest that employment status and repayment capacity should be considered during risk assessment.

6. Set Appropriate Loan Limits

The bank should consider applicants' income, existing debts, employment status, and ability to repay when determining loan amounts. Although loans above 30k had an 81% repayment rate compared with 80% for loans below 5k, the difference was small. Therefore, loan limits should be based on individual repayment capacity rather than loan size alone.